

$$\begin{aligned} & \frac{1-i_2}{16\pi^2} \left(\frac{1}{4} g_1^4 Y_e^{i1i2} LF_{2,2,-1}[m_1, m_e^{i2}] - \right. \\ & \frac{1}{32} g_1^2 Y_e^{i1i2} (g_1^2 + g_2^2) LF_{2,2,-1}[m_1, m_l^{i1}] + \frac{1}{32} g_2^2 Y_e^{i1i2} (-3g_1^2 + g_2^2) LF_{2,2,-1}[m_2, m_l^{i1}] + \\ & \frac{1}{4} g_1^2 Y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} LF_{2,2,0}[m_d^r, m_q^p] + \frac{1}{4} g_1^2 Y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} LF_{2,2,0}[m_e^r, m_l^p] - \\ & \frac{1}{4} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} LF_{2,1,0}[m_e^r, \tilde{\mu}] + \frac{1}{8} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} LF_{2,2,-1}[m_e^r, \tilde{\mu}] - \\ & \frac{1}{2} g_1^4 Y_e^{i1i2} LF_{2,1,0}[m_e^{i2}, m_1] - \frac{1}{2} g_2^2 Y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} LF_{3,1,0}[m_l^p, m_e^r] - \\ & \frac{1}{4} g_1^2 Y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} LF_{3,2,-1}[m_l^p, m_e^r] - \frac{3}{8} Y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} (g_1^2 - 3g_2^2) LF_{4,1,-1}[m_l^p, m_e^r] + \\ & \frac{1}{2} Y_e^{i1i2} \overline{a_e^{pr}} a_e^{pr} (g_1^2 - g_2^2) LF_{5,1,-2}[m_l^p, m_e^r] + \\ & \frac{1}{4} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} LF_{2,1,0}[m_l^p, \tilde{\mu}] - \frac{1}{8} g_1^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} LF_{2,2,-1}[m_l^p, \tilde{\mu}] + \\ & \frac{1}{16} g_1^2 Y_e^{i1i2} (g_1^2 + g_2^2) LF_{2,1,0}[m_l^{i1}, m_1] - \frac{1}{16} g_2^2 Y_e^{i1i2} (-3g_1^2 + g_2^2) LF_{2,1,0}[m_l^{i1}, m_2] + \\ & \frac{1}{2} Y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} (2g_1^2 - 3g_2^2) LF_{3,1,0}[m_q^p, m_d^r] - \frac{1}{4} g_1^2 Y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} LF_{3,2,-1}[m_q^p, m_d^r] - \\ & \frac{3}{8} Y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} (7g_1^2 - 9g_2^2) LF_{4,1,-1}[m_q^p, m_d^r] + \\ & \frac{3}{2} Y_e^{i1i2} \overline{a_d^{pr}} a_d^{pr} (g_1^2 - g_2^2) LF_{5,1,-2}[m_q^p, m_d^r] + \frac{1}{8} \tilde{\mu}^2 Y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 + 3g_2^2) \\ & LF_{2,2,0}[m_q^p, m_u^r] - \frac{1}{4} \tilde{\mu}^2 Y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 + 3g_2^2) LF_{3,1,0}[m_u^r, m_q^p] - \\ & \frac{1}{8} \tilde{\mu}^2 Y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 + 3g_2^2) LF_{3,2,-1}[m_u^r, m_q^p] - \frac{3}{4} \tilde{\mu}^2 Y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} \\ & (g_1^2 - 3g_2^2) LF_{4,1,-1}[m_u^r, m_q^p] + \frac{3}{2} \tilde{\mu}^2 Y_e^{i1i2} \overline{y_u^{pr}} y_u^{pr} (g_1^2 - g_2^2) LF_{5,1,-2}[m_u^r, m_q^p] + \\ & \frac{3}{2} \overline{y_d^{pr}} y_d^{sr} Y_e^{i1i2} \overline{y_u^{st}} y_u^{pt} LF_{2,1,0}[m_u^t, m_d^r] - \frac{3}{2} \overline{y_d^{pr}} y_d^{sr} Y_e^{i1i2} \overline{y_u^{st}} y_u^{pt} LF_{3,1,-1}[m_u^t, m_d^r] + \\ & \frac{1}{2} g_1^2 Y_e^{i1i2} (-g_1^2 + g_2^2) LF_{3,1,-1}[\tilde{\mu}, m_1] - \frac{3}{8} g_1^4 Y_e^{i1i2} LF_{3,2,-2}[\tilde{\mu}, m_1] - \\ & \frac{3}{8} g_1^4 m_1^2 Y_e^{i1i2} LF_{3,2,-1}[\tilde{\mu}, m_1] + g_1^2 Y_e^{i1i2} (g_1^2 - g_2^2) LF_{4,1,-2}[\tilde{\mu}, m_1] + \\ & \frac{1}{4} g_1^4 Y_e^{i1i2} LF_{4,2,-3}[\tilde{\mu}, m_1] + \frac{1}{4} g_1^4 m_1^2 Y_e^{i1i2} LF_{4,2,-2}[\tilde{\mu}, m_1] + \\ & \frac{1}{2} g_1^2 Y_e^{i1i2} (-g_1^2 + g_2^2) LF_{5,1,-3}[\tilde{\mu}, m_1] + \frac{3}{2} g_2^2 Y_e^{i1i2} (-g_1^2 + g_2^2) LF_{3,1,-1}[\tilde{\mu}, m_2] - \\ & \frac{15}{8} g_2^4 Y_e^{i1i2} LF_{3,2,-2}[\tilde{\mu}, m_2] - \frac{3}{8} g_2^4 m_2^2 Y_e^{i1i2} LF_{3,2,-1}[\tilde{\mu}, m_2] + \\ & 3g_2^2 Y_e^{i1i2} (g_1^2 - g_2^2) LF_{4,1,-2}[\tilde{\mu}, m_2] + \frac{5}{4} g_2^4 Y_e^{i1i2} LF_{4,2,-3}[\tilde{\mu}, m_2] + \\ & \frac{1}{4} g_2^4 Y_e^{i1i2} (m_2^2 + 2\tilde{\mu}^2) LF_{4,2,-2}[\tilde{\mu}, m_2] + \frac{3}{2} g_2^2 Y_e^{i1i2} (-g_1^2 + g_2^2) LF_{5,1,-3}[\tilde{\mu}, m_2] + \\ & \frac{1}{16} m_1 g_1^2 a_e^{i1i2} (-g_1^2 + g_2^2) LF_{2,2,1,-1}[m_1, m_e^{i2}, m_l^{i1}] - \\ & \frac{3}{4} g_1^4 Y_e^{i1i2} LF_{2,1,1,-1}[m_1, m_e^{i2}, \tilde{\mu}] - \frac{1}{2} g_1^4 m_1^2 Y_e^{i1i2} LF_{2,1,1,0}[m_1, m_e^{i2}, \tilde{\mu}] + \\ & \frac{1}{2} g_1^4 Y_e^{i1i2} LF_{3,1,1,-2}[m_1, m_e^{i2}, \tilde{\mu}] + \frac{1}{2} g_1^4 m_1^2 Y_e^{i1i2} LF_{3,1,1,-1}[m_1, m_e^{i2}, \tilde{\mu}] + \\ & \frac{1}{16} m_1 g_1^2 a_e^{i1i2} (-g_1^2 + g_2^2) LF_{2,2,1,-1}[m_1, m_l^{i1}, m_e^{i2}] - \frac{3}{16} g_1^4 Y_e^{i1i2} LF_{2,1,1,-1}[m_1, m_l^{i1}, \tilde{\mu}] - \\ & \frac{1}{8} g_1^4 m_1^2 Y_e^{i1i2} LF_{2,1,1,0}[m_1, m_l^{i1}, \tilde{\mu}] + \frac{1}{8} g_1^4 Y_e^{i1i2} LF_{3,1,1,-2}[m_1, m_l^{i1}, \tilde{\mu}] + \\ & \frac{1}{8} g_1^4 m_1^2 Y_e^{i1i2} LF_{3,1,1,-1}[m_1, m_l^{i1}, \tilde{\mu}] + \frac{1}{8} g_1^2 Y_e^{i1i2} (7g_1^2 - g_2^2) LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_e^{i2}] + \\ & \frac{3}{4} g_1^4 m_1^2 Y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_e^{i2}] + \frac{1}{8} g_1^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^p] + \\ & \frac{1}{16} g_1^2 Y_e^{i1i2} (-4g_1^2 + g_2^2) LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_l^{i1}] - \frac{3}{16} g_1^4 m_1^2 Y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_l^{i1}] - \\ & \frac{15}{16} g_2^4 Y_e^{i1i2} LF_{2,1,1,-1}[m_2, m_l^{i1}, \tilde{\mu}] - \frac{1}{8} g_2^4 m_2^2 Y_e^{i1i2} LF_{2,1,1,0}[m_2, m_l^{i1}, \tilde{\mu}] + \\ & \frac{5}{8} g_2^4 Y_e^{i1i2} LF_{3,1,1,-2}[m_2, m_l^{i1}, \tilde{\mu}] + \frac{1}{8} g_2^4 m_2^2 Y_e^{i1i2} LF_{3,1,1,-1}[m_2, m_l^{i1}, \tilde{\mu}] - \\ & 3g_1^2 g_2^2 \tilde{\mu}^2 Y_e^{i1i2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] + 2g_1^2 g_2^2 \tilde{\mu}^2 Y_e^{i1i2} LF_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \\ & \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_e^r] + \frac{3}{8} g_2^2 \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pi2} y_e^{i1r} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^p] + \\ & \frac{1}{16} g_2^2 Y_e^{i1i2} (3g_1^2 + 22g_2^2) LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_l^{i1}] + \frac{5}{16} g_2^4 m_2^2 Y_e^{i1i2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_l^{i1}] + \\ & \frac{1}{8} m_1 g_1^2 a_e^{i1i2} (5g_1^2 - g_2^2) LF_{2,1,1,0}[m_e^{i2}, m_1, m_l^{i1}] + \\ & \frac{1}{8} m_1 g_1^2 a_e^{i1i2} (-g_1^2 + g_2^2) LF_{3,1,1,-1}[m_e^{i2}, m_1, m_l^{i1}] + \\ & \frac{1}{2} g_1^4 Y_e^{i1i2} LF_{2,1,1,-1}[m_e^{i2}, m_1, \tilde{\mu}] - \frac{1}{8} m_1 g_1^2 a_e^{i1i2} (g_1^2 + 3g_2^2) LF_{2,1,1,0}[m_l^{i1}, m_1, m_e^{i2}] + \\ & \frac{1}{8} m_1 g_1^2 a_e^{i1i2} (-g_1^2 + g_2^2) LF_{3,1,1,-1}[m_l^{i1}, m_1, m_e^{i2}] + \frac{1}{8} g_1^2 Y_e$$